



Is Wi-Fi safe?

The WPA and WPA2 encryption has been broken by Russians using a brute force attack coupled with massively parallel computational power of NVIDIA's GPU

Did you know?

Modern day automobiles have more computational power than there was on the Apollo 13 moonlander

What goes on in your television set?

At a lower level, every embedded device is basically the same. It consists of:

- A microprocessor which is designed for low power consumption
- Memory – meant for storing both the original program ('firmware') and for processing of data
- A reliable, real-time Operating System or 'firmware' designed to give instant results while running without any errors for years at a time.

Perhaps the most significant use of the embedded microprocessor has been in making this world a safer place to live. A plane armed with an anti-collision device can warn the pilot if another plane comes too close. A modern ICU monitors all the vital signs of a patient and informs the medics should a parameter cross some dangerous threshold. The airbag controller in a car is so finely tuned that it can activate the safety bag before you hit the steering column. Most of the vital safety devices today contain some or the other form of computing support, because by nature, a computer cannot commit "human error".

The Spanair crash, which occurred on August 20th 2008 and cost 154 lives, is one of the examples of what happens when safety systems go wrong. This crash occurred because the pilots did not verify flap positions, and ground system that was supposed to verify this, was shut down due to a malware attack.

Another example is that of Therac-25, a radiation therapy machine which gave its patients severe burns and radiation poisoning. The reason was a software bug. This bug had been present in previous versions of the machines as well, but was never seen before due to mechanical locks. In the Therac-25 design, these locks were removed, but the software was not tested thoroughly.

These and many such instances depict how we have started placing the most precious thing in the hands of machines, human life. While casualties due to self-aware systems like the Skynet may still be in the future, a machine not following the proper safety standards has been around already. Hence the need for extensive safety norms like Do-178B, that ensure the

reliability and functionality of life-critical systems. With the increasing usage of embedded technology, we need these systems to be up and running, always.

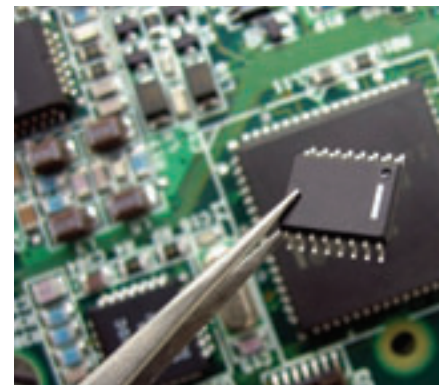
An aircraft while landing cannot afford the infamous blue screen of death (BSOD) during touchdown. A pacemaker planted inside a patient cannot afford to hang. That is why the operating systems which run inside these embedded devices are not standard operating systems. They are designed with special requirements like no hang issues, to always respond within a time limit, and to have a very high degree of reliability. The usual ones are QNX Neutrino (used in Cisco Equipment), VxWorks (one of the oldest), Integrity (used in USAF F-22 Raptor fighter jets), and its lighter version, the VelOSity. The most common open-source RTOS available is RTLinux.

Programming Embedded systems

- More on vehicle ECU modifications: http://en.wikipedia.org/wiki/Engine_control_unit#Programmable_ECUs
- <http://www.eetimes.com/design/embedded>
- Embedded Linux forum: <http://www.ucdot.org/>
- <http://www.embeddedarm.com/>

The most common hardware for embedded systems today are usually SOCs or System-On-Chip (example: Nvidia Tegra). This is a special chip which contains everything a system needs, including a CPU, memory, and other required controllers (graphics, input output, etc). The most common CPU used in such systems is either an ARM or an MIPS based CPU. Intel has recently released an Atom SoC as well. Unfortunately, most of the embedded systems available today cannot be reprogrammed without specialized equipment. This is both good and bad. Good because we may inadvertently change a mission-critical parameter. Bad because we won't be able to customize our equipment. While some ECUs (Electronic Control Unit) used in vehicles can be modified for aftermarket modifications, the best way is usually to swap out the complete unit.

Embedded devices already dwarf computers many times over in numbers, and



Advances in fabrication mean that the chips only get smaller

this is only going to increase as we move to a more and more automated future. With the rise of networking, the number of devices connected to the internet is increasing, so much so that this is acknowledged as a factor in the coming exhaustion of IP addresses, and the resulting move to IPv6 from the current IPv4. The day is not far off when most of the internet will consist of such devices talking to each other or sending updates on their status.

What was once considered the domain of some crazy programmers a few years ago is already here. You can, with some effort, put the technology in place to control your home appliances and other devices from a remote location. The smart home is not far away, when all such operations will be performed automatically using sensors to detect events, and react accordingly.

Cyborgs, once considered the fantasy of science fiction writers, are now a reality with humans increasingly being given artificial electronic implants to restore damaged body functions, from prosthetic implants to pacemakers, all with the help of embedded technology.

Embedded technology is definitely here to stay in our lives for the long term, but like most technologies, it is a double-edged sword. While making life easier for everyone involved and automating multiple tasks, it is still up to the designers to ensure that all such designs perform as expected. A good embedded design is always the one where the user does not have to think about the complexities working behind the scenes. 